Coauthor Dossier Ambitious

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Most Ambitious Version Of The Double-Entry Quantum Chapter

Purpose

This dossier is for coauthors, not for publication. It summarizes the package structure, the main claim decisions, the remaining checks, and the likely reviewer defenses. It is meant to make the ambitious version easier to discuss without reopening every sentence of the chapter.

Package Architecture

File	Intended reader	Function
Chapter.rewritten.ambitious.docx	Chapter readers and editors	Main publishable chapter; conceptually strong but not overloaded
TechnicalSupplement.ambitious.docx	Technical readers and coauthors	Formal notation, assumptions, propositions, proof sketches, source crosswalk
CoauthorDossier.ambitious.docx	Coauthors	Claim decisions, open checks, reviewer-response bank

The architecture deliberately separates exposition from verification. The main chapter should be readable as a chapter. The supplement should be checkable as technical support. The dossier should make coauthor decisions explicit.

Executive Judgment

The best version of the chapter is not "accounting is quantum." It is:

> *Double-entry accounting is a classical relational information architecture. Its power comes from preserving structured relationships among accounts, statements, and events. Quantum information theory gives a modern formal vocabulary for seeing why those relationships matter.*

This is stronger than a metaphor and weaker than a physical quantum claim. That middle position is the chapter's safest and most interesting territory.

Major Decisions Already Made

Decision	Rationale
Keep "quantum era" as context, not hype	Quantum information motivates the comparison, but the chapter is about accounting architecture
State universality as representational	The source supports representation of a universal gate set, not physical computation

Treat amplitudes as separate bookkeeping	Prevents the accounting notation from being mistaken for complete quantum probability
Make articulation the accounting center	The channel-capacity result is strongest when framed as articulation preserving distinctions
Keep odd-cycle result narrow	The result supports relational advantage in a specified model, not a general claim about all coordination
Move technical details to supplement	Lets the main chapter stay readable while keeping the formal defense available

Remaining Coauthor Checks

1. Confirm that the description of Fellingham, Lin, and Schroeder (2022) does not overstate the decision/accounting/information equivalence.
2. Confirm the Demski, Fellingham, Lin, and Schroeder (2008) performance-measurement summary before relying heavily on the agency implications.
3. Decide whether the terms "superposition property" and "entanglement property" should remain as named accounting properties or be softened throughout to "superposition analogue" and "entanglement analogue."
4. Confirm the n-account wording in the chapter and supplement: one-account/two-bit capacity is separate from the broader n-bit extension.
5. Decide whether the main chapter should keep all formal callouts or move some of them entirely into the supplement.

Claim Discipline Matrix

Claim	Status	Keep?	Wording guardrail
Double entry is relational information architecture	Interpretive synthesis	Yes	"Offers one explanation," not "fully explains history"
T-accounts can represent basis labels	Representation claim	Yes	"Represent," not "are"
Journal entries can represent gates	Representation claim	Yes	Amplitudes tracked separately
Toffoli plus Hadamard gives universality	Established theorem plus representation	Yes	"Can represent a universal gate set"
One account can transmit two bits under articulation	Source result	Yes	State scale abstraction and articulation conditions

More accounts automatically mean more bits	Overstatement	No	Require expanded message structure and articulation
Bell states model nonseparable relationships	Established result plus analogy	Yes	"Model" or "analogue," not literal accounts entangled
Odd-cycle entangled strategy beats classical	Source result	Yes	"In the specified three-seat model"
Superposition alone performs worse	Source result	Yes	"In this model," not generally
Group measurement can preserve joint information	Analogy plus cited agency source	Yes, cautiously	Verify 2008 source before strengthening

Reviewer-Response Bank

Objection: Are you saying accounting is quantum?

No. The chapter says double entry is a classical relational information architecture. Quantum information theory supplies a language for describing relationships, transformations, nonseparability, and measurement.

Objection: If amplitudes are external, what does accounting add?

The accounting notation adds a concrete representation of basis labels, transformations, conditional operations, and articulated relationships. It does not replace amplitude calculus; it makes the relational structure visible in accounting language.

Objection: Does representational universality mean accounting systems can compute quantum algorithms?

No. It means the notation can represent a universal gate set when paired with amplitude bookkeeping. Physical quantum computation requires physical qubits and coherent quantum dynamics.

Objection: Why abstract away from magnitudes in the channel-capacity model?

The abstraction isolates the information contribution of classification and articulation. With magnitudes included, the model would answer a different question about data plus structure rather than structure itself.

Objection: Why does an unchanged second account matter?

Because the receiver reads the manipulated account inside an articulated structure. The unchanged account preserves row relationships that prevent signals from collapsing under the model's equivalence rule.

Objection: Are the organizational implications proved?

No. The agency discussion is an analogy disciplined by the formal models and by cited performance-measurement theory. It identifies a research direction: measure at the level where payoff-relevant information is preserved.

Recommended Submission Strategy

Use the ambitious chapter as the candidate main text. Send the technical supplement to coauthors alongside it. Use this dossier for internal discussion and reviewer preparation, not as a submission artifact unless an editor asks for a response memo.

If page limits are tight, cut from the main chapter in this order:

6. Reduce the worked channel-capacity example table but keep the prose explanation.
7. Move one or two formal callouts entirely into the supplement.
8. Shorten the performance-measurement analogy.
9. Preserve the thesis, amplitude caveat, channel-capacity conditions, and odd-cycle numerical comparison.

Bottom Line

The ambitious version is bolder because it is more disciplined. It does not make the quantum claim louder. It makes the accounting claim clearer: articulation is a code for relational information.